Curriculum Vitae

PERSONAL INFORMATION

EMRAN ALI [Name: Emran Surname: Ali]

School of Information Technology,
Deakin University (Melbourne Burwood Campus), VIC 3125, Australia.

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Gender: Male | Nationality: Bangladeshi (By birth)

PERSONAL STATEMENT

I am an enthusiastic, hardworking, and patient life-long learner who has experience in teaching and research at university with industrial development skills. I developed excellent critical thinking, problem solving, computer programming, communication and management skills throughout my studies and professional periods. I wish to expand and use my knowledge and experiences to contribute to the outcome based productive progresses in leading universities, institutes and industries anywhere in the world.

TEACHING EXPERIENCE

(2) Sept. 2017 – Jan. 2022
(1) Sept. 2014 – Aug. 2017

Faculty Member (Assistant Professor[2], Lecturer[1]) – Currently on study leave

Department of Computer Science and Engineering (CSE)

Hajee Mohammad Danesh Science and Technology University (HSTU), Dinajpur-5200, Bangladesh. (http://hstu.ac.bd)

Responsibilities:

- Teaching: Theory and sessional course design, lecture material preparation, course conduction using lecturing, tutorial, demonstration, practical, workshop and seminar.
- Assessment: Exam question preparation, exam invigilation, answer script evaluation, practical experiment assessment, conducting viva voce, preparing and delivering results.
- Research: Research supervision, contribution in research and preparation for publication.

CASUAL TEACHING EXPERIENCE

(4) Jul. 2022 – Nov. 2022
(3) Feb. 2022 – Jun. 2022
(2) Jul. 2021 – Nov. 2021
(1) Jul. 2020 – Nov. 2020

Casual Academic [1,2,3,4]

School of Information Technology (IT), Faculty of Science, Technology and Build Environment (SEBE)

Deakin University, Melbourne Burwood Campus, Australia. (http://deakin.edu.au)

Responsibilities:

- SIT 720: Machine Learning and Python programming tutor, course conduction and exam preparation and assessment.
- Student HelpHub tutor. Machine Learning and Python programming, programming in C, C++, and Database.

CASUAL RESEARCH JOB EXPERIENCE

(2) Mar. 2022 – Till date
(1) Aug. 2021 – Dec. 2021

Casual Research Assistant

Deakin University, Melbourne Burwood Campus, Australia. (http://deakin.edu.au)

(The research and publication works had run for more than the official paid period. This project run from Jun. 2020 – Sep. 2022.[1])

Involvement:

- Collaboration with AETMOS, Australia and Monash University, Australia for air quality prediction (air temperature prediction). A MAAP linkage ARC project funded by the Australian Research Council (ARC).[2]
- Research and collaboration with Monash University, Australia and Alfred Health, Australia for the initial design of a seizure detection system for ICU patients.[1]

INDUSTRIAL WORK EXPERIENCE

May. 2012 – Aug. 2014

Software Development Manager and Senior Software Engineer

Advanced Apps Bangladesh Limited. (AAPBD) [Currently working with AppBajar and previously worked as DROIDBD], Dhaka, Bangladesh. (https://www.aapbd.com)

Responsibilities:

- iOS application development with web and social API.
- Project and team management: iOS, Android and Web team.
- Team management, client management, project management and application testing.

Feb. 2012 – May. 2012

Junior Programmer

Kento Studios Limited, Dhaka, Bangladesh. (http://www.kentostudios.com)

Responsibilities:

- iOS application development.
- Game design.

ACADEMIC INFORMATION

Jan. 2023 – Till Date (Continuing)

Doctor of Philosophy (PhD) in Information Technology (Deakin University)

Deakin University, Melbourne Burwood Campus, Australia. (http://deakin.edu.au)

Cotutelle (joint) with

Coventry University, Coventry Main Campus, the UK. (https://www.coventry.ac.uk)

Program Period: 3 years (maximum completion duration 4 years)

Funding: Deakin University Post-graduate Research Scholarship (DUPRS) and PGR Studentship during the stay at Coventry University.

Research and Thesis Title: A Machine Learning Based Analysis of Sleep Patterns in Aging Using Physiological Signals

Experiences and Activities:

- Research methodology
- Python programming language
- Digital signal processing
- Machine Learning (ML) and Deep Learning (DL) with Python
- Health Informatics
- Explainable/interpretable ML.
- Exploring the relationship between sleep and aging using Bio-signal processing, Machine Learning (Deep Learning) and interpretability.

Feb. 2020 – Aug. 2022
(Awarded in Oct. 2022)

Master of Science (by Research) in Information Technology — Awarded

Deakin University, Melbourne Burwood Campus, Australia. (http://deakin.edu.au)

Program Period: 2 years (maximum completion duration 3 years)

Funding: Bangabandhu Science and Technology Fellowship (BSTF). Funded by: Bangabandhu Science and Technology Fellowship Trust (BSTFT), The Ministry of Science and Technology, The Government of Peoples Republic of Bangladesh.

Research and Thesis Title: Minimising Electroencephalogram Channels for Epileptic Seizure Detection

Experiences and Activities:

- Research methodology
- Python programming language
- Digital signal processing
- Machine Learning (ML) and Deep Learning (DL) with Python
- Health Informatics
- Research and development in medical diagnosis and disease detection using Bio-signal processing, Machine Learning (Deep Learning) and optimization.

Jan. 2007 – Dec. 2010
(Awarded in Feb. 2012)

Bachelor of Science in Computer Science and Engineering (B.Sc in CSE) — A- (3.65 out of 4.0)

Hajee Mohammad Danesh Science and Technology University (HSTU), Dinajpur-5200, Bangladesh. (http://hstu.ac.bd)

Program Period: 4 years (maximum completion duration 7 years. During my studentship, entire university faced the delay of 1 year and 2 months for political problem in the country)

Research and Thesis Title: J2ME English to English dictionary application with improved binary search (Indexed-Binary search) technique. (Research is conducted on the last year of the program)

Core Courses:

- Core Mathematics, Discrete and Concrete Mathematics, Fundamental Physics
- Computer Programming, Databases, Software and System Development and Engineering
- Artificial Intelligence, Machine Learning, Pattern Recognition
- Hardware, Microprocessors, Digital System Design, Electronics and Digital Electronics
- Project and thesis in the last one year

RESEARCH INVOLVEMENT

Research Interest

Machine Learning and Deep Learning in the applied fields especially- healthcare, automation, biosignal analysis, sleep and aging.

- Data Science
- Machine Learning and Deep Learning
- Bio-signal Processing & Analysis
- Epilepsy, Sleep, Aging and Environment
- Biomedical Engineering & Medical InfoTech
- Disease Detection and Prediction
- Time-series Data Analysis
- Human-Computer Interaction

Research Publications (Posters)

- *"Deep Learning Model for Detection of Electrographic Seizures from continuous EEG in ICU patients"*, American Epilepsy Society (AES) Annual Conference 2021, 22 November 2021. Link: https://cms.aesnet.org/abstractslisting/deep-learning-model-for-detection-of-electrographic-seizures-from-continuous-eeg-in-icu-patients

Research Publications (Conferences)

- *"Performance Analysis of Entropy Methods in Detecting Epileptic Seizure from Surface Electroencephalograms,"*, IEEE Engineering in Medicine & Biology Society (EMBC), 2021, Maxico City, Maxico. Link: https://doi.org/10.1109/EMBC46164.2021.9629538c
- "*Efficient Noise Reduction and HOG Feature Extraction for Sign Language Recognition",* ICAEEE, 2018, Gazipur, Bangladesh. Link: https://doi.org/10.1109/ICAEEE.2018.8642983
- *"Indexed Binary Search based efficient search generator for J2ME English to English dictionary",* ICIEV, 2014, Dhaka, Bangladesh. Link: https://doi.org/10.1109/ICIEV.2014.6850692

Research Publications (Journals)

- *"A LSB Based Image Steganography Using Random Pixel and Bit Selection for High Payload,"*, IJMSC, 2021. Link: https://doi.org/10.5815/ijmsc.2021.03.03
- *"Enhancement of Single-Handed Bengali Sign Language Recognition Based on Hog Features."*, JATIT, 2020. Link: http://www.jatit.org/volumes/Vol98No5/2Vol98No5.pdf
- *"Mapping Character Position Based Cryptographic algorithm with Numerical Conversions."*, IJCSSE, 2020. Link: https://ijcsse.org/published/volume9/issue3/p1-V9I3.pdf
- "*Pseudo Random Ternary Sequence and Its Autocorrelation Property Over Finite Field",* IJCNIS, 2017. Link: https://doi.org/10.5815/ijcnis.2017.09.07
- "*Smart Campus Using IoT with Bangladesh Perspective: A Possibility and Limitation",* IJRASET, 2017. Link: https://doi.org/10.22214/IJRASET.2017.8239
- "*Study of Abstractive Text Summarization Techniques",* AJER, 2017. Link: https://www.ajer.org/papers/v6(08)/ZE0608253260.pdf
- "Performance Analysis of Energy Detection Based Spectrum Sensing Over Wireless Non-Fading AWGN Channels in Cognitive Radio Network", IJSER, 2017. Link: https://www.ijser.org/researchpaper/Performance-Analysis-of-Energy-Detection-Based-Spectrum-Sensing-Over-Wireless-Non-Fading-AWGN-Channels-in-Cognitive-Radio-Network.pdf

PERSONAL SKILLS

Language Skills

- Bengali: Native language
- English: Professional competence level.
  - IELTS: 6.5, (L,R,W,S-6.5, 6.0, 6.5, 7.0)
  - Medium of instruction and assessment at HSTU and Deakin University.

Programming Language Skills

- Python (Time-series analysis, Signal analysis, Machine Learning and Deep Learning)
- Objective-C (used in software development for iOS)
- Java, C, C++, C#, R language

Document Writing and Presentation Skills

- Latex and Microsoft Word (for word processing)
- Adobe Illustrator (for graphics design)
- Latex Beamer and Microsoft PowerPoint (for presentation)

ADDITIONAL INFORMATION

Scholarship and Fellowship

Scholarship:

- Deakin University Post-graduate Research Scholarship (DUPRS), Deakin University, Australia.

Fellowship:

- Bangabandhu Science and Technology Fellowship (BSTF), Bangabandhu Science and Technology Fellowship Trust (BSTFT), Ministry of Science and Technology, Bangladesh.
  Deakin University Post-graduate Research Scholarship (DUPRS), Deakin University, Australia.

Seminars

- Speaker, A seminar on Higher Studies in Abroad - Journey to Australia and Research Experiences, Organized by CSE Club, HSTU; September. 2022.
- Keynote Speaker, A seminar on *Career development and future of Mobile Application Development in Bangladesh*, Organized by CSE Association, HSTU; Apr. 2015.

Projects and Funding

Research Projects:

- Research Supervisor, *Development of three-layer network architecture to prevent web servers from DDoS attack*, Funded by IRT (Institute of Research and Training), HSTU; Jul. 2019 – Jun. 2020.
- Research Supervisor, *Analysis and design of a non-invasive way of measuring, monitoring and predicting blood-sodium concentration level using near-infrared spectroscopy*, Funded by IRT (Institute of Research and Training), HSTU; Jul. 2019 – Jun. 2020.

Honours and Awards

- Cocutelle PhD Fellow, Deakin University, Australia & Coventry University, the UK, 2023.
- Best Presentation Award, Annual Conference (2021), School of IT, Deakin University.
- Best employee award in AAPBD, 2013.
- Ranked 1st from Bangladesh in the entrance exam-2013 for Masters Program in South Asian University (SAU), India, operated by SAARC (https://www.saarc-sec.org).
- Ranked 1st in the batch in Bachelors from CSE department at HSTU, 2012.
- Ranked 1st in the Intra-University Computer Programming Contest (CPC) at HSTU, 2009.

Memberships

- Member, Deakin University Student Association (DUSA)
- Member, Deakin University AI Society (DAIS)
- Member, Deakin University Software Engineering Club
- IEEE student member.
- IEEE EMBS (Engineering in Medicine and Biology Society) student member.
- Advisor and founding member, HSTU-Programmers Arena, HSTU
- Advisor, CSE Club of HSTU, HSTU

Social Networks

- LinkedIn: https://bd.linkedin.com/in/wwmemran
- Google Scholar: https://scholar.google.com/citations?user=ej8d6LkAAAAJ
- Research Gate: https://www.researchgate.net/profile/Emran_Ali
- GitHub: https://github.com/WWM-EMRAN
- ORCiD: https://orcid.org/0000-0003-3095-5265

REFERENCES

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Senior Lecturer
School of Information Technology,
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